

```
$ mysql -e "EXPLAIN SELECT * FROM users WHERE email = ?"
```



//  $O(n)$   $O(\log n)$

**B-Tree**

□□□

**EXPLAIN**

□□□□

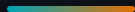
□□□□

□□□□□□

□□

□□□□□

\$ cat index-plan.md



01 /



vs



02 /



B-Tree EXPLAIN



03 /



04 /



100 3000ms 3ms

## PART 01



1000 000000 Apple

□□□□□□□□

```
1  □□□□
2  [1] → [2] → [3] → ... → [1000000]
3
4  □□□email = 'hua@email.com'
5  遍历 1 遍历
```

EXPLAIN

type: ALL□□□□□□

rows: 1000000

🕒  $O(n)$  - □□□□□□□□

□□□□□□□□

```
1  遍历      root
2           /      \
3         branch    branch
4           /      \
5        leaf      leaf → □□□□
```

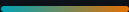
EXPLAIN

type: ref□□□□□□

rows: 1

🕒  $O(\log n)$  - 3~4 次 I/O □□

□□□□□□□□□□



| □□□□      | □□□     | □□□ | □□□□          |
|-----------|---------|-----|---------------|
| 1,000     | 2ms     | 1ms | 2x            |
| 10,000    | 25ms    | 1ms | 25x           |
| 100,000   | 300ms   | 2ms | 150x          |
| 1,000,000 | 3,500ms | 3ms | <b>1000x+</b> |

**O(n)** □□□□


 ×10
 
 ×10

**O(log n)** □□□□


 ×10
 

000<1000 00000000000>10 0000000000

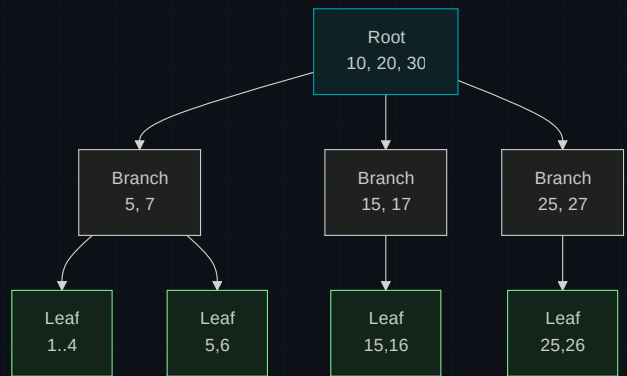
PART 02

# B-Tree □□

□□□□ 3~4 □ I/O□

# B-Tree

NO GRAPH



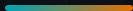
□□  
□□□□□□□□

□□  
□□□□□□□□

□□□□  
□□□□□□□□

□□□  
□□□□I/O □□

□□□□□□□□□□



```
1  -- 创建索引
2  CREATE INDEX idx_user_email ON users(email);
3
4  -- 查询
5  SELECT * FROM users WHERE email = 'hua@email.com';
6  -- type: ALL, rows: 1000000 → type: ref, rows: 1
```

□□□□□□□□



□□□□□

□□□□PRIMARY KEY □□□□□□□□□□□□

\$ 创建索引 EXPLAIN 类型 ref / range 扫描 ALL 扫描

## PART 03



□□□□□

```
1  -- WHERE □□
2  CREATE INDEX idx_order_status
3      ON orders(status);
4
5  -- JOIN □□□
6  CREATE INDEX idx_order_user_id
7      ON orders(user_id);
8
9  -- ORDER BY □□
10 CREATE INDEX idx_order_created
11     ON orders(created_at);
```

□□□□□□

□□□□□□  
gender □□□□□□□□□□□□

□□□□□□□  
last\_login\_time □□□□□□□□□□

□□□□□□□  
description □□□□□□□□□□□□



```
1  -- 100 rows
2  SELECT * FROM orders
3  WHERE user_id = 123 AND status = 'completed'
4         AND order_date >= '2024-01-01';
5
6  -- Create index
7  CREATE INDEX idx_orders_user_status_date
8         ON orders(user_id, status, order_date);
9  -- rows: 1000000 → 1200 8000
```

**1. user\_id**

1000000

**2. status**

1000000

**3. order\_date**

1000000

## PART 04



0000 ≠ 0000

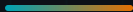
CHECKPOINT\_01

name 姓 B-Tree 索引

A WHERE UPPER(name) = 'ZHANG'

B WHERE name LIKE '%姓'

C WHERE name LIKE '姓%'



```
1 SELECT * FROM users WHERE UPPER(name) = 'ZHANG'; -- ✗ 不区分大小写
2 SELECT * FROM users WHERE name = 'Zhang'; -- ✓ 区分大小写
3
4 SELECT * FROM users WHERE name LIKE 'Z%'; -- ✗ 模糊查询
5 SELECT * FROM users WHERE name LIKE 'Z%'; -- ✓ 模糊查询
6
7 SELECT * FROM users WHERE age + 10 > 35; -- ✗ 算术运算
8 SELECT * FROM users WHERE age > 25; -- ✓ 比较运算符
```

| SQL 语句                                    | SQL 语句                                    | SQL 语句                                    |
|-------------------------------------------|-------------------------------------------|-------------------------------------------|
| SELECT * FROM users WHERE name = 'Zhang'; | SELECT * FROM users WHERE name = 'Zhang'; | SELECT * FROM users WHERE name = 'Zhang'; |
| SELECT * FROM users WHERE name LIKE 'Z%'; | SELECT * FROM users WHERE name LIKE 'Z%'; | SELECT * FROM users WHERE name LIKE 'Z%'; |
| SELECT * FROM users WHERE age + 10 > 35;  | SELECT * FROM users WHERE age + 10 > 35;  | SELECT * FROM users WHERE age + 10 > 35;  |
| SELECT * FROM users WHERE age > 25;       | SELECT * FROM users WHERE age > 25;       | SELECT * FROM users WHERE age > 25;       |



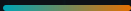
#### GAIN

- ✓ 時間  $O(n) \rightarrow O(\log n)$
- ✓ ORDER BY / GROUP BY 高速
- ✓ JOIN 高速

#### COST

- × 実行時間 10-30%増
- × INSERT / UPDATE / DELETE 高速
- × メモリ = 実行時間

80/20 法則 20% の実行時間削減を実現



□□□□□□□□ — □□□□□□□□□□

# Query optimized.

0000000000 - 00000 Hash 00

```
$ EXPLAIN SELECT ... -- type: ref, rows: 1  
b-tree → explain → composite → 0000
```